Acta Oceanol. Sin., 2016, Vol. 35, No. 4, P. 63–73
DOI: 10.1007/s13131-016-0822-0
http://www.hyxb.org.cn
E-mail: hyxbe@263.net

Implementing a multispecies size-spectrum model in a data-poor ecosystem

ZHANG Chongliang[1, 2], CHEN Yong[2], THOMPSON Katherine[2], REN Yiping[1*]

[1] College of Fisheries, Ocean University of China, Qingdao 266003, China

[2] School of Marine Sciences, University of Maine, Orono, Maine 04469, USA

Received 19 May 2015; accepted 28 August 2015

©The Chinese Society of Oceanography and Springer-Verlag Berlin Heidelberg 2016

Abstract

Multispecies ecological models have been used for predicting the effects of fishing activity and evaluating the performance of management strategies. Size-spectrum models are one type of physiologically-structured ecological model that provide a feasible approach to describing fish communities in terms of individual dietary variation and ontogenetic niche shift. Despite the potential of ecological models in improving our understanding of ecosystems, their application is usually limited for data-poor fisheries. As a first step in implementing ecosystem-based fisheries management (EBFM), this study built a size-spectrum model for the fish community in the Haizhou Bay, China. We describe data collection procedures and model parameterization to facilitate the implementation of such size-spectrum models for future studies of data-poor ecosystems. The effects of fishing on the ecosystem were exemplified with a range of fishing effort and were monitored with a set of ecological indicators. Total community biomass, biodiversity index, W-statistic, LFI (Large fish index), MeanW (mean body weight) and Slope (slope of community size spectra) showed a strong non-linear pattern in response to fishing pressure, and largest fishing effort did not generate the most drastic responses in certain scenarios. We emphasize the value and feasibility of developing size-spectrum models to capture ecological dynamics and suggest limitations as well as potential for model improvement. This study aims to promote a wide use of this type of model in support of EBFM.

Key words: size-spectrum model, trophic interaction, data-poor, model parameterization, Haizhou Bay

Citation: Zhang Chongliang, Chen Yong, Thompson Katherine, Ren Yiping. 2016. Implementing a multispecies size-spectrum model in a data-poor ecosystem. Acta Oceanologica Sinica, 35(4): 63–73, doi: 10.1007/s13131-016-0822-0

1 Introduction

Multispecies ecological models have been developed for predicting the effects of fishing activity and evaluating performance of management strategies since ecosystem-based fisheries management (EBFM) is widely recognized (Plagányi, 2007; Smith et al., 2015). Ecological models are useful tools for addressing complex ecological processes (McCook, 1994; Bolker, 2008; Schmolke et al., 2010; Sillero, 2011), such as trophic interactions that primarily shape the structure and dynamics of marine ecosystems (Sissenwine, 1984; Hollowed et al., 2000; Hunsicker et al., 2011). However, trophic interactions are complex in marine ecosystems because fishes are commonly opportunistic predators, thus most predator-prey relationships are weakly coupled (Hunsicker et al., 2011). Furthermore, food preferences of individual fish change with growth, a phenomenon referred to as "life-history omnivory" (Werner and Gilliam, 1984; Pimm and Rice, 1987). Consequently, variations in trophic relationships, such as prey switching, ontogenetic diet shifts and cannibalism, commonly occur in marine ecosystems and are often overlooked by traditional predator-prey models such as Lotka–Volterra equations (Jennings et al., 2001).

Ideally, a model should address ontogenetic shifts in competitive and/or predatory relationships among species, instead of assuming a fixed interaction (Hartvig and Andersen, 2013), and size spectra provide one promising approach. Body size was shown to play a fundamental role in determining predator-prey interactions (Sheldon et al., 1982; Peters, 1983; Dickie et al., 1987; Pope et al., 2006; Rossberg, 2012). It was suggested that individual body size is closely related to trophic level and is more important in defining an organism's trophic niche than species identity (Denney et al., 2002; Jennings et al., 2002). In addition, body size was used to delineate biomass spectrum, i.e., aggregated biomass is relatively constant or slightly declines as a function of body size (Sheldon et al., 1972; Dickie et al., 1987; Boudreau and Dickie, 1992). These features suggest that it is feasible to describe the structure of marine ecosystems based on size spectra (Andersen and Beyer, 2006; Hartvig et al., 2011).

Size-spectrum models comprise one category of physiologically-structured process models (De Roos and Persson, 2001; Hartvig and Andersen, 2013). This type of model provides a conceptually simple way to describe predator–prey interactions with regard to the complexity of individual diet variation and ontogenetic trophic shift (Persson et al., 2000; Hartvig and Andersen, 2013; Svanbäck et al., 2015). The size-spectrum model addresses two central processes underlying fish population dynamics including (1) food-dependent growth and (2) size-dependent predation (Hartvig et al., 2011; Persson et al., 2014). The model formulates biological processes at the individual level and applies

Foundation item: The Special Fund for Agriscientific Research in the Public Interest under contract No. 201303050; the Fundamental Research Funds for the Central Universities under contract Nos 201022001 and 201262004.

*Corresponding author, E-mail: renyip@ouc.edu.cn

metabolic theory to relate process rates to individual body weight (Brown et al., 2004). This allows the model to be simply structured and to have few parameters (Pope et al., 2006; Hartvig et al., 2011). The model can be used to build a simulation framework with various applications. This type of model is being increasingly utilized to explore food web dynamics (Silvert and Platt, 1978; Benoît and Rochet, 2004; Law et al., 2009; Hartvig et al., 2011) and to evaluate the effects of fishing activity and management strategies (Andersen and Pedersen, 2010; Blanchard et al., 2014; Andersen and Beyer, 2015).

Despite the important role of ecological models in improving our understanding of ecosystem dynamics, most ecological models have been designed for well-studied marine ecosystems whereas application to data-poor stocks is limited (Mills et al., 2011; Costello et al., 2012; Patrick and Link, 2015). It is often not feasible to obtain sufficient data at the time scale needed for modeling due to limited public attentions and finances, not to mention that data may never be sufficient to understand the full complexity of ecosystems (Levin, 2000; Browman and Stergiou, 2004).

This study built a size-spectrum model for the fish community in the Haizhou Bay, China to exemplify the use of ecological models for a data-poor ecosystem. The Haizhou Bay is a poorly studied marine region with a long history of fisheries exploitation (Wang et al., 1993). The bay used to support a productive fishery, however, the production has dropped dramatically since the end of last century (Tang et al., 2011), and the reason for the decline is largely unidentified. This study provides an example of modeling an insufficiently studied ecosystem. Although model building procedures were introduced and well established (e.g., Houle et al., 2013; Blanchard et al., 2014), modeling details pose great challenges for a poorly-studied ecosystem. Data collection procedures and model parameterization were explained in this study and simulations were run with the model to provide examples of potential applications. Finally, we identify limitations of the model and suggest the potential for improvement. This study aims to promote a wide use of size-spectrum models for exploring the ecosystems dynamics in the context of EBFM.

2 Materials and methods

2.1 *Model description*

The size-spectrum model was developed by Andersen and Beyer (2006), Andersen and Pedersen (2010) and Hartvig et al. (2011) and is available as the R package "Multi-species sIZE spectrum modelling in R (mizer)" (Scott et al., 2014a). The "mizer" model incorporates biological processes such as growth, reproduction, mortality and metabolism at the level of an individual with a set of sub-models (Table 1). This model considers food consumption and energy flow within the food web, assuming that the energy used for metabolism, growth and reproduction comes from consumption of prey. The allometric scaling law was used to relate biological process rates to body size of individual organisms. The multispecies population dynamics were characterized by the size-structured McKendrick–von Foerster equation (McKendrick, 1925; von Foerster, 1959):

$$\frac{\partial N(w)}{\partial t}+\frac{\partial g(w)N(w)}{\partial w}=-\mu(w)N(w),$$

where $N(w)$ is the fish abundance and the growth g and mortality μ are species-specific and depend on body weight (w). Sub-models of the main biological processes and their general parameters are described below and summarized in Tables 1 and 2 respectively.

(1) Individual food preference is determined by the body size ratio between predator and prey, which follows a log-normal function (Ursin, 1973):

$$\phi\left(\frac{w_{\mathrm{p}}}{w}\right)=\exp\left[\frac{-\left(\ln\left(w/\left(w_{\mathrm{p}}\beta\right)\right)\right)^2}{2\sigma^2}\right],$$

where w is individual body weight of a predator, w_{p} is the body weight of a prey, β is the preferred species-specific size ratio, and σ is the width of the size selection function.

(2) Food sources derived from fish community itself and background resources including plankton and other small organism. The growth of background resources is assumed to follow a logistic-type function with a constant instinct rate $r_0 w^{p-1}$and carrying capacity $\kappa_{\mathrm{R}} w^{-\lambda}$ that can be estimated for a given ecosystem (Hartvig et al., 2011):

$$\frac{\partial N_{\mathrm{R}}(w)}{\partial t}=r_0 w^{p-1}\left[\kappa_{\mathrm{R}} w^{-\lambda}-N_{\mathrm{R}}(w)\right]-\mu_{\mathrm{r}}(w)N_{\mathrm{R}}(w),$$

where N_{R} is the resource density, and p and λ are scaling exponents listed in Table 2. The predator-prey encounter rate is described by the "Andersen-Ursin" encounter model (Andersen and Ursin, 1977):

$$E_i(w)=V(w)\int\left(N_{\mathrm{R}}(w_{\mathrm{p}})+\sum_j\theta_{ij}N_j(w_{\mathrm{p}})\right)\phi(w_{\mathrm{p}}/w)\,w_{\mathrm{p}}\mathrm{d}w_{\mathrm{p}}.$$

Encounter rate is the product of a weight-dependent food searching rate $V(w)$ (defined in Table 2) and the density of available food defined with the integration. Available food is determined by fish abundance, background resources $\left(N_{\mathrm{R}}(w_{\mathrm{p}})+\sum_j\theta_{ij}N_j(w_{\mathrm{p}})\right)$, and the size selection function. The θ represents the strength of an interspecific interaction attributed to species-specific preference or overlap in distribution.

(3) Growth (g) and reproduction (g_{r}) depend on food consumption and individual maturity:

$$g(w)=(\alpha f(w)\,I_{\max}-k_{\mathrm{s}}w^p)(1-\psi(w)),$$

$$g_{\mathrm{r}}(w)=(\alpha f(w)\,I_{\max}-k_{\mathrm{s}}w^p)\,\psi(w).$$

The consumed food $f(w)I_{\max}$ (feeding level and maximum consumption rate, defined in Table 1) is assimilated with efficiency α, and the obtained energy is firstly used for the consumption of metabolism ($k_{\mathrm{s}}w^p$) (Hartvig et al., 2011). The remaining energy is divided between growth and reproduction depending on the maturity of individuals (ψ, defined in Table 1).

(4) The amount of energy allocated to reproduction determines egg production (Hartvig et al., 2011):

$$R_{\mathrm{p}}=\frac{\varepsilon}{2w_0}\int N(w)\,g_{\mathrm{r}}(w)\,\mathrm{d}w,$$

where w_0 is the egg size, the factor 1/2 assumes a 1:1 sex ratio. The reproductive efficiency ε represents the efficiency of converting energy into offspring. Recruitment is assumed to be density-

Table 1. Sub-model equations of the multispecies size-spectrum model, modified from Blanchard et al. (2014)

Process	Sub model	Equation
Predation	prey size selection	$\phi\left(\frac{w_\mathrm{p}}{w}\right)=\exp\left[\frac{-\left(\ln\left(w/\left(w_\mathrm{p}\beta\right)\right)\right)^2}{2\sigma^2}\right]$
	volumetric search rate	$V(w)=\frac{f_0 h\beta^{2-\lambda}}{(1-f_0)\sqrt{2\pi}\kappa\sigma}w^q$
	encountered food	$E(w)=V(w)\sum_i\theta_i\int N_i(w)\phi\left(\frac{w_\mathrm{p}}{w}\right)w_\mathrm{p}\mathrm{d}w_\mathrm{p}$
	feeding level	$f(w)=\frac{E(w)}{E(w)+hw^n}$
Growth	maturation	$\psi(w)=\left[1+\left(\frac{w}{w_\mathrm{mat}}\right)^{-10}\right]^{-1}\left(\frac{w}{W}\right)^{1-n}$
	somatic growth	$g(w)=\left(\alpha f(w)I_\mathrm{max}-k_\mathrm{s}w^p\right)(1-\psi(w))$
Recruitment	energy for reproduction	$g_\mathrm{r}(w)=\left(\alpha f(w)I_\mathrm{max}-k_\mathrm{s}w^p\right)\psi(w)$
	egg production	$R_\mathrm{ep}=\frac{\varepsilon}{2w_0}\int N(w)g_\mathrm{r}(w)\mathrm{d}w$
	recruitments	$R=R_\mathrm{max}\frac{R_\mathrm{ep}}{R_\mathrm{ep}+R_\mathrm{max}}$
Mortality	background mortality	$\mu_0=Z_0W^{n-1}$
	predation mortality	$\mu_\mathrm{p}(w_\mathrm{p})=\sum_i\int\phi_i\left(\frac{w_\mathrm{p}}{w}\right)\left(\left(1-f_i(w)\right)V_i(w)\theta_iN_i(w)\mathrm{d}w\right.$
	starvation mortality	$\mu_\mathrm{s}=\begin{cases}0 & k_\mathrm{s}w^p<\alpha f(w)hw^n\\ \frac{k_\mathrm{s}w^p-\alpha f(w)hw^n}{\xi w} & \text{otherwise}\end{cases}$
Resources	resource dynamics	$\frac{\partial N_\mathrm{R}(w)}{\partial t}=r_0w^{n-1}\left[\kappa(w)-N_\mathrm{R}(w)\right]-\mu_{\mathrm{p.r}}(w)N_\mathrm{R}(w)$
	carrying capacity	$\kappa(w)=\kappa_\mathrm{r}w^{-\lambda}$

Notes: In the equations, w is fish body weight, and the subscript p indicates prey species. The parameters used in the models are listed in Table 2 and Table 3.

dependent due to the compensatory nature of egg production and is described by a Beverton-Holt-type model in our study (Beverton and Holt, 1957):

$$R=R_\mathrm{max}\frac{R_\mathrm{p}}{R_\mathrm{p}+R_\mathrm{max}}.$$

where the maximum potential recruitment (R_max) needs to be estimated for each species.

(5) Mortality is attributed to four sources: fishing, predation, starvation, and background mortality. Predation mortality is calculated as the sum of individuals consumed by each species (j) of all body size (Hartvig et al., 2011):

$$\mu_i(w_\mathrm{p})=\sum_j\int\phi_i\left(\frac{w_\mathrm{p}}{w}\right)\left(\left(1-f_j(w)\right)V_j(w)\theta_{ij}N_j(w)\mathrm{d}w.\right.$$

Starvation mortality occurs when food supply does not meet metabolic requirements (Hartvig et al., 2011):

$$\mu_\mathrm{s}=\begin{cases}0 & k_\mathrm{s}w^p<\alpha f(w)hw^n,\\ \frac{k_\mathrm{s}w^p-\alpha f(w)hw^n}{\xi w} & \text{otherwise,}\end{cases}$$

Background mortality is assumed to occur throughout an individual's lifetime (Hartvig et al., 2011):

$$\mu_0=Z_0W^{n-1}.$$

2.2 *Model parameterization*

The "mizer" package parameterizes species at three levels with increasing complexity (Scott et al., 2014b). At the first level, the community size-spectrum model characterizes all species in a fish community as being the same (Benoît and Rochet, 2004). At the next level of complexity, the trait-based size-spectrum model characterizes a species with asymptotic body size (Andersen and Beyer, 2006). Lastly, the multispecies size-spectrum model characterizes species with a set of life history parameters (Hartvig et al., 2011).

This study parameterized the multispecies size-spectrum model for the marine fish community in the Haizhou Bay, China as an example of data-poor region. The Haizhou Bay ranges from 5 to 40 m in depth and is located on the continental shelf adjacent to the Shandong Peninsula in China (Fig. 1). The hydrological condition of the bay has significant seasonal variations driven by the monsoon. Historically, the bay has served as an important nursery and spawning ground for many commercially-important fish species and has supported productive fisheries (Wang et al., 1993). The ecological dynamics of the bay have been poorly studied, however, large changes in species composition and biomass have been observed since the decline of local fisheries (Tang et al., 2011). As a result, trawlers have left the bay in search of fish and stow-net fisheries are responsible for the majority of fish production in the Haizhou Bay (Qi, 2013).

A series of surveys were conducted to collect data on the fish community and fishing activity in the Haizhou Bay for model parameterization. A stratified random bottom trawl survey with strata defined by depth and latitude was conducted in March, May, July, September and December 2011 (Fig. 1). There were 24 stations sampled each month to evaluate quantity and composition of the fish community. An otter trawl with 17 mm codend mesh was towed at the speed of 2 knot for 30 min in each

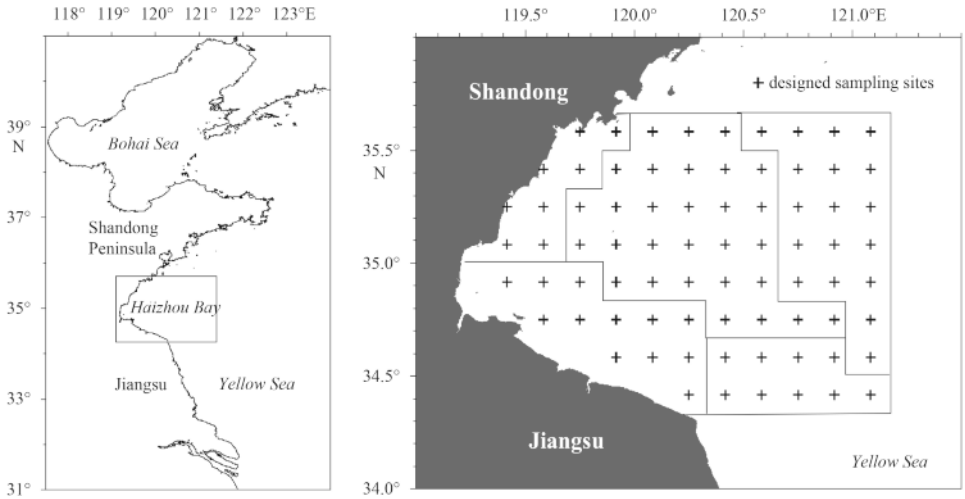


Fig. 1. Station locations (+) and strata (boundaries defined by lines) of the stratified survey in the Haizhou Bay, 2011.

sampling station. Over 300 species were identified in the four-season trawl survey, including 96 fishes, 29 shrimps, 38 crabs, and 84 mollusks. There is an obvious seasonal variation in the species composition of the community. The catch was sorted and weighted by species to estimate fish biomass. Fish measurement, stomach content analysis and age identification were preformed when the samples were sufficient.

The 23 most common species that accounted for over 90% of total fish biomass were included in the size-spectrum model (Table 3). Growth, predation, mortality, and reproduction were estimated for each species from the following data sources:

(1) The body-shape parameters, a and b in the weight-length relationship were estimated from trawl survey data. Growth parameters including infinite body weight (W_{inf}) and growth rate (k_{vb}) were estimated from the von Bertalanffy growth equation when age-length data were available. For species lacking growth information, data were obtained from FishBase (www.fishbase.org) and past studies that described fish biology in similar areas. The parameter estimates were averaged when multiple sources were available.

(2) Preferred predator–prey size ratio (β) was estimated from stomach content analysis as well as past studies. The width of the selection function was set to 1.3 since precise estimates were not available for most species (Andersen and Pedersen, 2010; Jacobsen et al., 2014).

(3) Size at maturity (W_{mat}) was obtained from related studies and 0.25× asymptotic body size was assumed for species without maturity information (Beverton, 1992; Andersen and Pedersen, 2010; Houle et al., 2012).

(4) It was assumed that fish select prey based upon body size and no species preferences. A species interaction matrix was used to account for spatial and temporal overlaps in species' distributions (Blanchard et al., 2014). The matrix was estimated with an overlap index calculated from the trawl survey data:

$$\theta_{ij} = 1 - \frac{1}{2}\sum |p_i - p_j| ,$$

where p_i is the proportion of biomass of species i in the fish community. Intraspecific predation could not be properly estimated with the available data and were constrained to 0 based on the lack of cannibalismobserved in the stomach content analysis of the trawl survey.

(5) Physiological parameters, including the maximum consumption rate, standard metabolism coefficient and assimilation efficiency were based on the default model settings (Table 2, Houle et al., 2012; Jacobsen et al., 2014).

(6) Fishing parameters were estimated from the stow-net survey in the Haizhou Bay (Qi, 2013). Fishing mortality in the size-spectrum model is a product of three components: fishing effort, size selectivity and catchability. Stow-net selectivity was described by a "knife-edge" function that assigned each species a critical body size above which individuals were vulnerable to the net (Scott et al., 2014b). The critical body size was estimated from weight composition of the stow-net survey data for each species. The "knife-edge" function was a practical approach to addressing lack of species-specific gear selectivities. Preliminary analysis showed that this simplification provided consistent results compared with a range of sigmoid selectivity functions.

Catchability was used to describe the availability of fish to capture by fishing gears regardless of size. Catchability is rarely addressed in fishery surveys as it is highly dependent on gear design, fish behavior (e.g., response to gear, vertical distribution and swimming endurance), as well as spatial and temporal distribution of fishes and fishing gears (Fraser et al., 2007). This study evaluated the catchability of each species in stow-nets by comparing the total weight of stow-net landings and the estimated biomass from the trawl survey. We assumed that the fish composition observed in the trawl survey was a proper representation of the real fish community. Accordingly, the ratio between stow-net landings in the fishery survey (Qi, 2013) and total biomass estimated from the trawl survey using sweeping area method provided an approximation of catch rate, which was equal to the product of fishing effort and catchability when size selectivity was equal to 1 in the "knife-edge" function.

Fishing effort (which is different from fishing mortality in this study) has not been assessed in the Haizhou Bay in the past. We assumed that the stow-net fishery imposed the same fishing effort on all species and the differences in catch rates arose from catchability and size selectivity. The catchability of the species with the highest catch rate (yellow croaker, *Pseudosciaena polyactis*) was calibrated to 1 to confine the range of catchability (0, 1) and fishing effort and catchability for other species were derived accordingly. Although the estimated fishing effort and catchability are relative, the calibration did not affect model predictions.

(7) The carrying capacity of background resources (κ_{R}) and

Table 2. Summary of general parameters in the size-spectrum model

Symbol	Parameter	Value	Unit
Growth			
f_0	initial feeding level	0.6	–
h	coefficient of maximum food intake	$h = \frac{3K\nu b}{af_0} W^{1/3}$	$g^{1-n}.a^{-1}$
n	exponent for maximum food intake	2/3	–
σ	width of size selection function	1.3	–
q	exponent for volumetric search rate	0.8	–
α	assimilation efficient	0.6	–
p	exponent of standard metabolism	0.7	–
λ	exponent of resource spectrum	$2-n+q$	–
k_s	coefficient of standard metabolism	$0.2\,h$	$g^{1-n}.a^{-1}$
Resources			
κ	carrying capacity of the background recourses	1.01×10^9	$g^{\lambda-1}.Vol^{-1}$
r_0	coefficient of regeneration rate of resources	10	$g^{1-n}.a^{-1}$
w_{cut}	upper weight limit of the resource spectrum	10	g
Reproduction			
W_0	offspring weight	1.0	mg
ε	efficiency of offspring production	0.1	–
Mortality			
z_0	factor for background mortality	0.6	–
S	selectivity with knife-edge function, equal 1 if individual body size is larger than a threshold (SS) and 0 otherwise	$S(w) = (0,1)$	–
F	fishing mortality, determined as the product of C, species-specific catchability, S, size-selectivity and E, fishing efforts	$F(w,i) = S(w)\,C(i)\,E$	a^{-1}

maximum recruitment (R_{max}) were calibrated using an optimal approach to fit the model prediction to the observed species composition (Blanchard et al., 2014). This approach calculated the divergence of the model prediction from the observed community structure in log scale and used the "optim" algorithm in the R package "stats" to minimize the sum of squared errors. The "L-BFGS-B" method was used to constrain parameter estimation to a reasonable range by setting lower and upper bounds for each variable (Byrd et al., 1995). The estimate in Blanchard et al. (2014) was used as the initial value of R_{max} and the total biomass of organisms weighing less than five grams in the trawl survey was used to estimate the initial value of κ_R required for the "optim" algorithm.

Since the model simulated ecological processes such as the growth and reproduction rate on the scale of a year, the model prediction was considered an annual average. In order to stabilize the fitting process with limited observations averaged over a year, the model was run for 100 years and the prediction of the last 20 years was used to calculate divergence in model calibration. This approach assumed that the fish community was in a state of equilibrium over long-term simulation runs (Houle et al., 2012). The species-specific parameters are summarized in Table 3.

2.3 *Simulation*

The calibrated model was used for 100 years to equilibrium with the estimated fishing effort, and the prediction at equilibrium was compared to the observed fish biomass in the trawl survey. A range of fishing effort (0, 0.5, 1, 2 and 3) was simulated to exemplify the effects of fishing pressure on the fish community. The model was first run for 40 years without fishing pressure, following which we imposed gradually increasing fishing effort for 10 years and then fishing effort was stabilized for the remaining 50 years. The state of the fish community was monitored with a set of commonly used ecological indicators, including total biomass (B), biodiversity index (H'), W statistic based on an abundance-biomass comparison curve (*W-statistic*), large fish index (*LFI*), mean body weight of the fish community (*meanW*) and community size spectrum slope (*Slope*), for which the definition and calculation methods were summarized in Table 4. The abundance and biomass used for deriving ecological indicators only accounted for individuals vulnerable for catch on the trawl survey.

3 Results

3.1 *Basic output*

The default model setting defined 100 size-bins for the fish community spectrum and 130 bins for background resources on a logarithmic scale (Scott et al., 2014b). The model projection provides a "MizerSim" objective (a S4 objective in R), in which the slot "*n*" records the size-specific abundance of each species over time (i.e., *n* is a three dimensional array of time × species × size-bin). A variety of outputs can be derived based on the data (Fig. 2). The biomass of most species reached equilibrium after 50 years (Fig. 2a) with constant fishing effort. The biomass spectra exhibited the size composition of the fish community and background resources (the top line) and showed a sharp decrease of fish abundance at large body sizes (Fig. 2b). Predation mortality varied among species and generally decreased with size except for troughs around 0.1 g and 10 g (Fig. 2c). Fishing mortality varied among species as a result of the species-specific catchability (Fig. 2d).

The predicted population biomass matched observations on the trawl survey for most species with the exception of three (Fig. 3). Tanaka's snailfish (*Liparis tanakae*) and octopus (*Octopus ocellatus*) were considerably underestimated, while Thrissa kammalensis (*Kammal thryssa*) were overestimated (Fig. 3).

3.2 *Simulation of fishing effects*

The effect of fishing on community dynamics was monitored

Table 3. Summary of species-specific parameters in the "mizer" model

Species	Common name	W_{inf}/g	a	b	W_{mat}/g	β	Kvb/a^{-1}	R_{max}/a^{-1}	SS/g	C
Konosirus punctatus	dotted gizzard shad	256	0.008	2.99	27	2 730	0.27	3.2E8	8.41	0.88
Thrissa kammalensis	kammal thryssa	30	0.003	3.35	8	409	0.33	6.8E9	0.97	0.25
Hexagrammos otakii	fat greenling	2 400	0.014	3.00	50	84	0.23	5.5E7	2.00	0.06
Octopus ocellatus	octopus	252	0.017	2.52	63	56	1.40	7.6E6	1.83	0.02
Pholis fangi	gunnel	30	0.007	3.18	5	1 551	0.65	1.2E9	5.89	0.15
Paralichthys olivaceus	bastard halibut	15 706	0.012	3.00	506	47	0.15	2.2E5	–	0.00
Lophius litulon	yellow goosefish	13 496	0.016	3.10	260	88	0.35	5.3E4	2.47	0.02
Collichthys lucidus	big head croaker	78	0.024	2.82	16	24	0.42	1.3E8	–	0.00
Syngnathus acus	greater pipefish	7	0.000	3.67	0.4	570	1.00	6.1E9	0.11	0.85
Pleuronichthys cornutus	ridged-eye flounder	333	0.045	2.89	113	147	0.20	7.9E7	–	0.00
Amblychaeturichthys hexanema	pinkgray goby	31	0.013	2.89	4	220	0.70	4.1E9	0.77	0.96
Chaeturichthys stigmatias	finespot goby	82	0.010	3.01	9	758	0.70	9.0E7	0.77	0.58
Miichthys miiuy	miiuy croaker	2 913	0.009	3.09	342	33	0.32	1.4E5	–	0.00
Loligo sp.	squid	16	0.003	3.00	6	14	1.10	3.8E8	0.67	0.19
Liparis tanakae	tanaka's snailfish	2 307	0.023	2.81	31	14	0.65	4.5E5	2.84	0.03
Apogonichthys lineatus	Indian perch	21	0.020	3.18	6	2 609	0.80	2.2E7	0.43	0.01
Callionymus kitaharae	dragonet	59	0.009	3.05	8	602	0.52	4.6E8	0.44	0.53
Pseudosciaena polyactis	yellow croaker	579	0.027	2.81	31	312	0.50	2.6E8	0.91	1.00

Notes: W_{inf} and Kvb are the asymptotic body weight and growth rate in von Bertalanffy growth equation; a and b are the parameters in weight-length relationship (cm, g); W_{mat} represents the mature body size; β is preference ratio of body weight between predator and prey; R_{max} is the maximum recruitments in recruitment function, and SS is the selective size in the knife-edge selective function; C is the relative catching efficient of stow-net across species estimated from survey data.

Table 4. Summary of ecological indicators that characterize fish community structure

Indicators	Symbol	Definition	Calculation
Total biomass	*Biomass*	the sum of biomass of all individuals (Murawski, 2000).	use "getBiomass" in "mizer", and sum the biomass of all species
Shannon-Wiener diversity index	H'	$H' = -\sum p_i \ln p_i$, p_i indicates the biomass proportion of species i in fish community (Shannon, 1948).	use "diversity" in "vegan" on predited biomass
W statistic	*W-statistic*	the area between biomass and abundance cumulative curves across all species, $W = \sum \frac{B_{cum} - N_{cum}}{50\,(S-1)}$, B_{cum} and N_{cum} is the cumulative biomass and abundance respectively, and S is the total number of species (Clarke and Warwick, 1994).	use "getBiomass" and "getN" to access the biomass and abundance, and conduct the formulation.
Large fish indicator	*LFI*	the total biomass of all individuals with the body weight >100 g divided by total biomass of the community (Greenstreet et al., 2011).	"getProportionOfLargeFish" in "mizer"
Mean body weight	*meanW*	mean body weight of all individuals in the community (Rochet and Trenkel, 2003).	"getMeanWeight" in "mizer"
Slope of community size spectrum	*Beta*	$\log N_w = \alpha + \beta \log W$ the slope of a linear regression of log (abundance in each size class, N_w) versus log (body weight) (Rice and Gislason, 1996).	"getCommunitySlope" in "mizer"

with multiple ecological indicators (Fig. 4). The total community biomass decreased with increasing fishing effort (Fig. 4). Biodiversity index increased rapidly when fishing effort was imposed, whereas the index remained constant or decreased slightly when fishing effort was held constant (Fig. 4). This response might be attributed to the fish community composition being more even when abundant species are removed. A decrease in the W-statistic at higher fishing efforts suggests the removal of large species by the stow-net. It should be noted that the greatest fishing effort (F = 3) did not generate the most drastic responses. LFI decreased sharply in response to fishing, but increased after 10 years with fishing efforts larger than 2. *MeanW* also decreased with fishing pressure and then exceeded values resulting from lack of fishing after about 15 years. The *Slope* increased when $F \leqslant 2$, but decreased abruptly at F = 3. These results suggest a strong non-linear pattern in fish community dynamics in response to fishing pressure.

4 Discussion

There are multiple benefits of using multispecies models compared to the traditional single-species approach, including an improved understanding of variability in growth rates, natural mortality and recruitment dynamics (Hollowed et al., 2000; Fogarty, 2014). The most fundamental advantage of multispecies models is the ability to address indirect trophic effects with regard to energy flow within a food web (Pimm and Rice, 1987; Hollowed et al., 2000; Whipple et al., 2000). Size-spectrum models have some advantages over the other models that assume a fixed rate of predation, growth or recruitment such as the EwE model (Pauly et al., 2000; Christensen and Walters, 2004), the Atlantis model (Fulton et al., 2005, 2011), and the length-based model (Hall et al., 2006). The dynamics of this model emerged from the rate of underlying processes rather than fixed parameters (Scott

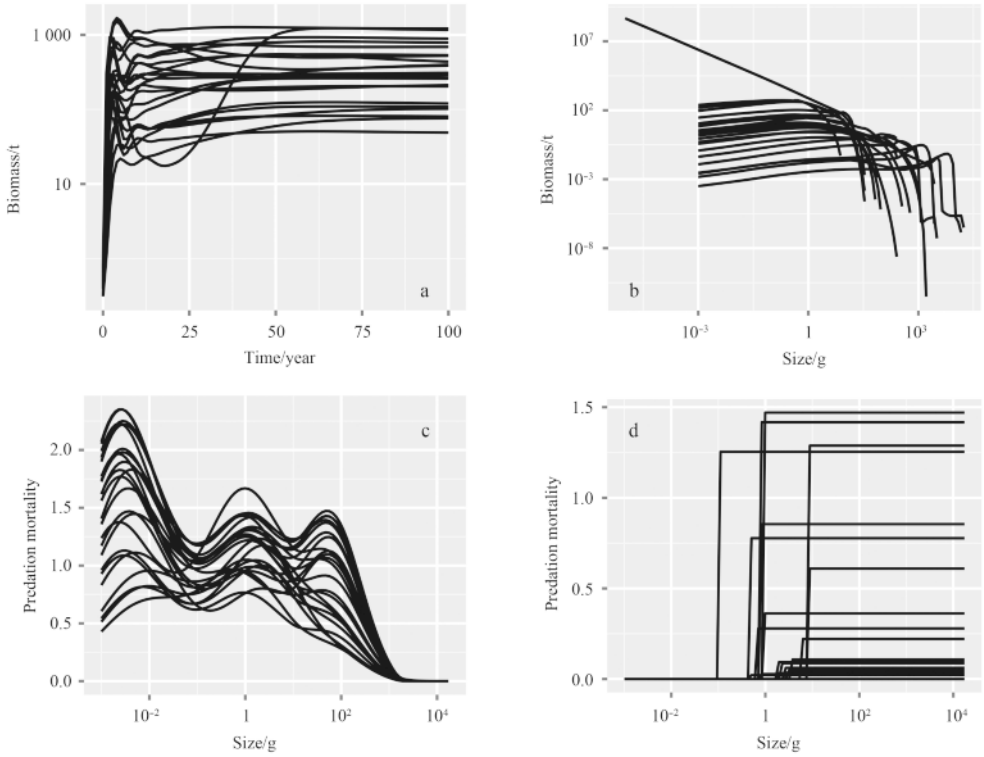


Fig. 2. Output summary of "mize" model on species biomass through time (a), the community biomass spectra (b), predation mortality (c) and fishing mortality (d) in equilibrium. Each line denotes one of the 23 species included in the model.

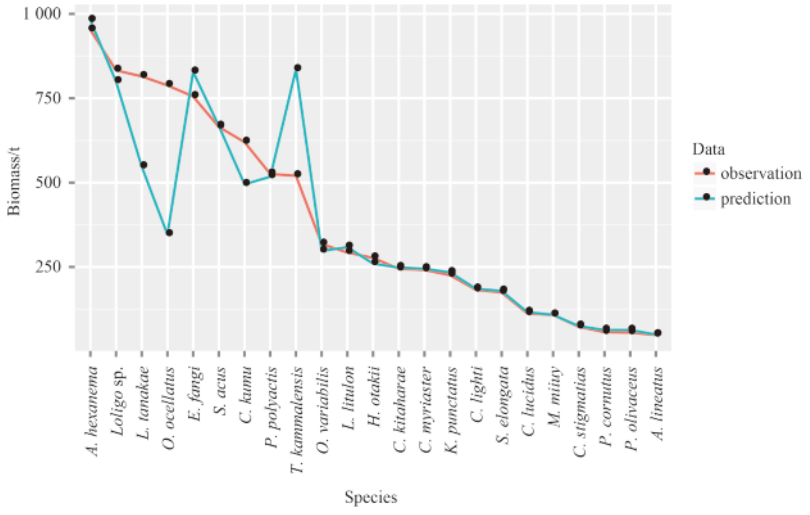


Fig. 3. A comparison of fish biomass between trawl survey observations and size-spectrum model predictions. The predicted biomass is an average over 10 years in equilibrium.

et al., 2014a). This model also explicitly addressed ontogenetic trophic niche shifts, which are critical in early life history (Jennings et al., 2001). These characteristics permit nonlinear population dynamics, illustrated both by our results as well as other related studies (Houle et al., 2012). The nonlinearity allows this model to simulate the complex trophodynamics of ecosystems in response to fishing pressure.

This study presents a practical perspective on model implementation (c.f., "essentially, all models are wrong, but some are useful", Box and Draper (1987)), however, it is worth highlighting some shortcomings of the model that need to be improved in future studies. Allometric scaling including the maximum food intake exponent (n) and the standard metabolic scaling exponent (p) vary among taxa (Hartvig et al., 2011) and the variability was shown to significantly influence model outputs (Zhang et al., 2015). The detrital food web was not considered in the model framework, which is thought to play a critical role in trophic structure, biodiversity and stability of ecosystems (Mann, 1988; Moore et al., 2004). Besides simplifications in model structure, several weaknesses should be noted regarding model parameterization: (1) this model included 23 of the most abundant species and excluded rare species. The exclusion of rare species might

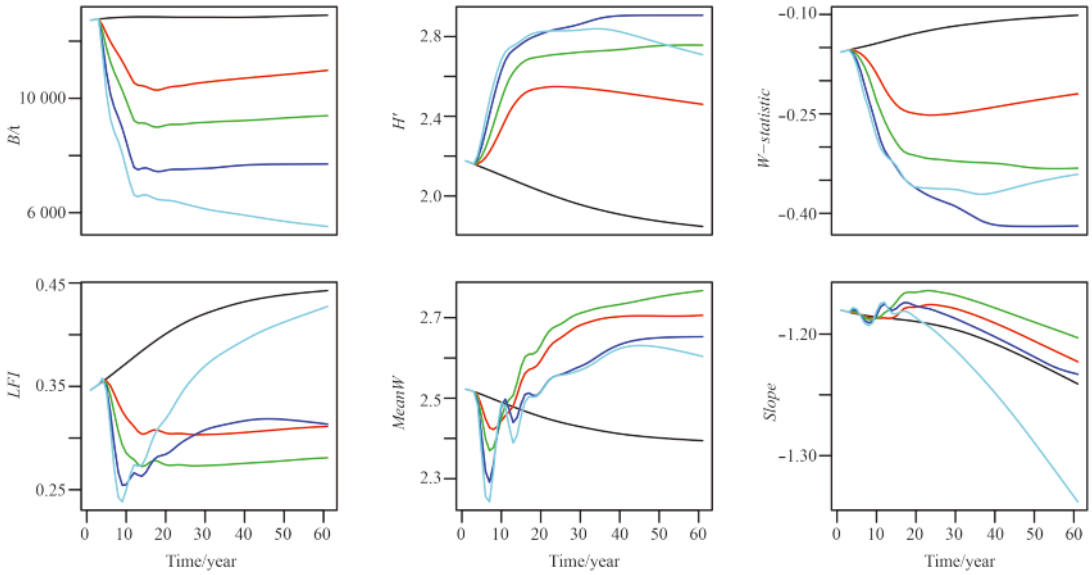


Fig. 4. Dynamics of ecological indicators in response to fishing effort. The indicators included total biomass (*B*), biodiversity index (*H'*), W statistic based on abundance-biomass comparison curve (*W-statistic*), large fish index (*LFI*), mean body weight of the fish community (*meanW*) and community size spectrum slope (*Slope*). The lines in black, red, green, blue and light blue denote the scenarios of $F = 0$, 0.5, 1.0, 2.0, 3.0, respectively.

lead to a biased projection of biodiversity and implicitly ignore the potential of rare species to become abundant in an ecosystem regime shift (Levin and Mollmann, 2014). (2) Spatial boundaries of the ecosystem were not considered in the simulation. The Haizhou Bay is not a closed system and migration and meta-population dynamics might influence the structure of the fish community (Ying et al., 2011). (3) Seasonality was ignored as the model was formulated on an annual basis, but temperate water and other environmental factors vary seasonally. Since reproductive processes occur on seasonal scale, annual projections may produce biased recruitment estimates, which consequently influence trophic interactions, population dynamics and the risk of population collapse.

Regarding both advantages and disadvantages of size-spectrum models, a general rule for application in fisheries management is that fisheries models should focus on the specific management problem and the concerns of stakeholders (Collie et al., 2016; Dickey-Collas et al., 2014). It has been argued that models can only achieve two attributes out of generality, realism, and precision (Dickey-Collas et al., 2014). This size-spectrum model as well as most other ecological models aims to achieve generality and realism over precision in order to gain general insight into ecological processes (Bolker, 2008; Smith et al., 2015). In other words, this model can not provide a precise assessment for ecosystems and statistical rather than mechanistic models are suggested to be more suitable for short-term projections (Schindler and Hilborn, 2015). On the other hand, comprehensive data and models may not be necessary depending on the goals of fisheries management (Patrick and Link, 2015), since qualitative models can sometimes effectively inform precautionary management strategies (Murawski, 2007; Hobday et al., 2011). A simple preliminary model that identifies ecological driving forces can be useful to efficiently prioritize research efforts in the early stages of EBFM (Patrick and Link, 2015). The proper use of minimal data could be valuable to answering ecological questions that a preliminary model can address. Retrospective analyses can be helpful for improving a model with additional data as they become available.

Ecosystem models can be categorized into three groups in terms of their objectives: (1) conceptual models that contribute to a general understanding of ecosystem process, (2) strategic models that provide information for strategic decision making, and (3) tactical models that provide short-term management advice (FAO, 2008). The size-spectrum model should be considered a strategic model, which can be used for understanding ecosystem function and processes (Smith et al., 2015). This model and many other ecological models provide a feasible framework for conducting ecosystem-scale experiments (Scheffer and Carpenter, 2003; Scott et al., 2014a), and the information derived from such a simulation could be helpful for guiding fisheries management (Collie et al., 2016). For example, (1) ecological models can be compared to traditional assessments to predict undesirable management consequences (Livingston et al., 2005), which facilitates a precautionary approach for management. (2) The prediction of ecological models helps to estimate key parameters of single-species models, such as natural mortality and recruitment, which used to be difficult to obtain or were highly variable (Garrison et al., 2010). (3) The projected dynamics of ecosystem can be used to evaluate the feasibility of management objectives (Andersen et al., 2015). (4) Ecological models help to identify key factors determining ecosystem dynamics and guide the development of tactical models (Collie et al., 2016). Many studies have been conducted to explore these topics (e.g., Law et al., 2009; Andersen and Pedersen, 2010; Hartvig et al., 2011; Blanchard et al., 2014; Andersen and Beyer, 2015). However, more widespread implementation of ecological models could incorporate ecological considerations into fisheries management processes (Garcia and Cochrane, 2005; Kruse et al., 2012). Studies examining ecosystem dynamics in the future should address environmental variability, spatial and temporal scales, and economic and social dimensions of fisheries (Jennings et al., 2001).

In conclusion, we highlight the value and feasibility of developing size-spectrum models to simulate ecosystems even in data-poor regions. We also propose several suggestions that may

improve future model development: (1) the spatial overlap index is subject to sampling errors and the use of a species distribution model (SDM, Stockwell and Peterson, 2002) may improve estimation of species interactions. (2) Species-specific preference for food and environmental conditions should be addressed when such data are available, as these traits influence the trophodynamics of ecosystems. (3) The background natural mortality should be adapted to fish age and longevity. (4) Seasonal variation in life history and environmental conditions could be included in the model to address seasonal dynamics that may have important implications for fisheries management actions such as seasonal closures. (5) Life history parameters and spatial distribution of larvae and juveniles should be addressed in the model, since age-specific traits and distribution can be critical for estimating mortality and recruitment.

Acknowledgements

The senior author's study in Chen's laboratory at the University of Maine is supported by the China Scholarship Council, Ocean University of China, and University of Maine.

References

Andersen K H, Beyer J E. 2006. Asymptotic size determines species abundance in the marine size spectrum. The American naturalist, 168(1): 54–61

Andersen K H, Beyer J E. 2015. Size structure, not metabolic scaling rules, determines fisheries reference points. Fish and Fisheries, 16(1): 1–12

Andersen K H, Brander K, Ravn-Jonsen L. 2015. Trade-offs between objectives for ecosystem management of fisheries. Ecological Applications, 25(5): 1390–1396

Andersen K H, Pedersen M. 2010. Damped trophic cascades driven by fishing in model marine ecosystems. Proceedings of the Royal Society B: Biological Sciences, 277(1682): 795–802

Andersen K P, Ursin E. 1977. A multispecies extension to the Beverton and Holt theory of fishing: with accounts of phosphorous circulation and primary production. Medd Dan Fisk Havunders, 7: 319–345

Benoît E, Rochet M J. 2004. A continuous model of biomass size spectra governed by predation and the effects of fishing on them. Journal of Theoretical Biology, 226(1): 9–21

Beverton R J H. 1992. Patterns of reproductive strategy parameters in some marine teleost fishes. Journal of Fish Biology, 41: 137–160

Beverton R J H, Holt S J. 1957. On the Dynamics of Exploited Fish Populations. Fisheries Investigation Series Ⅱ, XIX. London: Ministry of Agriculture, Fisheries and Food, UK, 533

Blanchard J L, Andersen K H, Scott F, et al. 2014. Evaluating targets and trade-offs among fisheries and conservation objectives using a multispecies size spectrum model. Journal of Applied Ecology, 51(3): 612–622

Bolker B M. 2008. Ecological Models and Data in R. Princeton: Princeton University Press, 408

Boudreau P R, Dickie L M. 1992. Biomass spectra of aquatic ecosystems in relation to fisheries yield. Canadian Journal of Fisheries and Aquatic Sciences, 49(8): 1528–1538

Box G E P, Draper N R. 1987. Empirical Model-Building and Response Surfaces. New York: John Wiley & Sons, 669

Browman H I, Stergiou K I. 2004. Perspectives on ecosystem-based approaches to the management of marine resources. Marine Ecology Progress Series, 274: 269–303

Brown J H, Gillooly J F, Allen A P, et al. 2004. Toward a metabolic theory of ecology. Ecology, 85(7): 1771–1789

Byrd R H, Lu P H, Nocedal J, et al. 1995. A limited memory algorithm for bound constrained optimization. SIAM Journal on Scientific Computing, 16(5): 1190–1208

Christensen V, Walters C J. 2004. Ecopath with Ecosim: Methods, capabilities and limitations. Ecological Modelling, 172(2–4): 109–139

Clarke K R, Warwick R M. 1994. Change in Marine Communities: An Approach to Statistical Analysis and Interpretation. Plymouth, UK: Plymouth Marine Laboratory, 172

Collie S, Botsford L W, Hastings A, et al. 2016. Ecosystem models for fisheries management: finding the sweet spot. Fish and Fisheries, 17(1): 101–125

Costello C, Ovando D, Hilborn R, et al. 2012. Status and solutions for the world's unassessed fisheries. Science, 338(6106): 517–520

de Roos A M, Persson L. 2001. Physiologically structured models - from versatile technique to ecological theory. Oikos, 94(1): 51–71

Denney N H, Jennings S, Reynolds J D. 2002. Life–history correlates of maximum population growth rates in marine fishes. Proceedings of the Royal Society of London. Series B: Biological Sciences, 269(1506): 2229–2237

Dickey-Collas M, Payne M R, Trenkel V M, et al. 2014. Hazard warning: model misuse ahead. ICES Journal of Marine Science, 71(8): 2300–2306

Dickie L M, Keer S R, Boudreau P R. 1987. Size-dependent processes underlying regularities in ecosystem structure. Ecological Monographs, 57(3): 233–250

FAO. 2008. Fisheries management. 2. The ecosystem approach to fisheries. 2. 1 Best practices in ecosystem modelling for informing an ecosystem approach to fisheries. FAO Fisheries Technical Guidelines for Responsible Fisheries. No. 4, Suppl. 2, Add. 1. Rome: FAO, 78

Fogarty M J. 2014. The art of ecosystem-based fishery management. Canadian Journal of Fisheries and Aquatic Sciences, 71(3): 479–490

Fraser H M, Greenstreet S P R, Piet G J. 2007. Taking account of catchability in groundfish survey trawls: implications for estimating demersal fish biomass. ICES Journal of Marine Science, 64(9): 1800–1819

Fulton E A, Link J S, Kaplan I C, et al. 2011. Lessons in modelling and management of marine ecosystems: the Atlantis experience. Fish and Fisheries, 12(2): 171–188

Fulton E A, Smith A D M, Punt A E. 2005. Which ecological indicators can robustly detect effects of fishing?. ICES Journal of Marine Science, 62(3): 540–551

Garcia S M, Cochrane K L. 2005. Ecosystem approach to fisheries: a review of implementation guidelines. ICES Journal of Marine Science, 62(3): 311–318

Garrison L P, Link J S, Kilduff D P, et al. 2010. An expansion of the MSVPA approach for quantifying predator-prey interactions in exploited fish communities. ICES Journal of Marine Science, 67(5): 856–870

Greenstreet S P R, Rogers S I, Rice J C, et al. 2011. Development of the EcoQO for the North Sea fish community. ICES Journal of Marine Science, 68(1): 1–11

Hall S J, Collie J S, Duplisea D E, et al. 2006. A length-based multispecies model for evaluating community responses to fishing. Canadian Journal of Fisheries and Aquatic Sciences, 63(6): 1344–1359

Hartvig M, Andersen K H. 2013. Coexistence of structured populations with size-based prey selection. Theoretical Population Biology, 89: 24–33

Hartvig M, Andersen K H, Beyer J E. 2011. Food web framework for size-structured populations. Journal of Theoretical Biology, 272(1): 113–122

Hobday A J, Smith A D M, Stobutzki I C, et al. 2011. Ecological risk assessment for the effects of fishing. Fisheries Research, 108(2–3): 372–384

Hollowed A B, Bax N, Beamish R, et al. 2000. Are multispecies models an improvement on single-species models for measuring fishing impacts on marine ecosystems?. ICES Journal of Marine Science, 57(3): 707–719

Houle J E, Andersen K H, Farnsworth K D, et al. 2013. Emerging asymmetric interactions between forage and predator fisheries impose management trade-offs. Journal of Fish Biology, 83(4): 890–904

Houle J E, Farnsworth K D, Rossberg A G, et al. 2012. Assessing the

sensitivity and specificity of fish community indicators to management action. Canadian Journal of Fisheries and Aquatic Sciences, 69(6): 1065–1079

Hunsicker M E, Ciannelli L, Bailey K M, et al. 2011. Functional responses and scaling in predator-prey interactions of marine fishes: contemporary issues and emerging concepts. Ecology Letters, 14(12): 1288–1299

Jacobsen N S, Gislason H, Andersen K H. 2014. The consequences of balanced harvesting of fish communities. Proceedings of the Royal Society B: Biological Sciences, 281(1775): 20132701

Jennings S, Greenstreet S R, Hill L, et al. 2002. Long-term trends in the trophic structure of the North Sea fish community: evidence from stable-isotope analysis, size-spectra and community metrics. Marine Biology, 141(6): 1085–1097

Jennings S, Kaiser M, Reynolds J D. 2001. Marine Fisheries Ecology. Oxford: Wiley-Blackwell, 432

Kruse G H, Browman H I, Cochrane K L, et al. 2012. Global Progress in Ecosystem-Based Fisheries Management. Alaska Sea Grant: University of Alaska Fairbanks, 386

Law R, Plank M J, James A, et al. 2009. Size-spectra dynamics from stochastic predation and growth of individuals. Ecology, 90(3): 802–811

Levin P S, Mollmann C. 2014. Marine ecosystem regime shifts: challenges and opportunities for ecosystem-based management. Philosophical Transactions of the Royal Society B: Biological Sciences, 370(1659): 20130275

Levin S A. 2000. Fragile Dominion: Complexity and the Commons. Reading, Massachusetts: Perseus Books, 250

Livingston P A, Aydin K, Boldt J, et al. 2005. A framework for ecosystem impacts assessment using an indicator approach. ICES Journal of Marine Science, 62(3): 592–597

Mann K H. 1988. Production and use of detritus in various freshwater, estuarine, and coastal marine ecosystems. Limnology and Oceanography, 33(4Part2): 910–930

McCook L J. 1994. Understanding ecological community succession: causal models and theories, a review. Vegetatio, 110(2): 115–147

McKendrick A G. 1925. Applications of mathematics to medical problems. Proceedings of the Edinburgh Mathematical Society, 44: 98–130

Mills D J, Westlund L, de Graaf G, et al. 2011. Under-reported and undervalued: small-scale fisheries in the developing world. In: Pomeroy R S, Andrew N, eds. Small-Scale Fisheries Management: Frameworks and Approaches for the Developing World. UK: CABI, 1–15

Moore J C, Berlow E L, Coleman D C, et al. 2004. Detritus, trophic dynamics and biodiversity. Ecology Letters, 7(7): 584–600

Murawski S A. 2000. Definitions of overfishing from an ecosystem perspective. ICES Journal of Marine Science, 57(3): 649–658

Murawski S A. 2007. Ten myths concerning ecosystem approaches to marine resource management. Marine Policy, 31(6): 681–690

Patrick W S, Link J S. 2015. Myths that continue to impede progress in ecosystem-based fisheries management. Fisheries, 40(4): 155–160

Pauly D, Christensen V, Walters C. 2000. Ecopath, Ecosim, and Ecospace as tools for evaluating ecosystem impact of fisheries. ICES Journal of Marine Science, 57(3): 697–706

Persson L, Byström P, Wahlström E. 2000. Cannibalism and competition in Eurasian perch: population dynamics of an ontogenetic omnivore. Ecology, 81(4): 1058–1071

Persson L, van Leeuwen A, de Roos A M. 2014. The ecological foundation for ecosystem-based management of fisheries: mechanistic linkages between the individual-, population-, and community-level dynamics. ICES Journal of Marine Science, 71(8): 2268–2280

Peters R H. 1983. The Ecological Implications of Body Size. Cambridge: Cambridge University Press, 329

Pimm S L, Rice J C. 1987. The dynamics of multispecies, multi-life-stage models of aquatic food webs. Theoretical Population Biology, 32(3): 303–325

Plagányi E E. 2007. Models for an ecosystem approach to fisheries. FAO Fisheries Technical Paper No. 477. Rome: FAO, 108

Pope J G, Rice J C, Daan N, et al. 2006. Modelling an exploited marine fish community with 15 parameters: results from a simple size-based model. ICES Journal of Marine Science, 63(6): 1029–1044

Qi Guangrui. 2013. Catch composition and effects on fish resources of the set-net along the coastal water of Haizhou Bay [dissertation]. Qingdao: Ocean Univerisy of China (in Chinese), 57

Rice J, Gislason H. 1996. Patterns of change in the size spectra of numbers and diversity of the North Sea fish assemblage, as reflected in surveys and models. ICES Journal of Marine Science, 53(6): 1214–1225

Rochet M J, Trenkel V M. 2003. Which community indicators can measure the impact of fishing? A review and proposals. Canadian Journal of Fisheries and Aquatic Sciences, 60(1): 86–99

Rossberg A G. 2012. A complete analytic theory for structure and dynamics of populations and communities spanning wide ranges in body size. Advances in Ecological Research, 46: 427–521

Scheffer M, Carpenter S R. 2003. Catastrophic regime shifts in ecosystems: linking theory to observation. Trends in Ecology & Evolution, 18(12): 648–656

Schindler D E, Hilborn R. 2015. Prediction, precaution, and policy under global change. Science, 347(6225): 953–954

Schmolke A, Thorbek P, DeAngelis D L, et al. 2010. Ecological models supporting environmental decision making: a strategy for the future. Trends in Ecology & Evolution, 25(8): 479–486

Scott F, Blanchard J L, Andersen K H. 2014a. *mizer*: an R package for multispecies, trait-based and community size spectrum ecological modelling. Methods in Ecology and Evolution, 5(10): 1121–1125

Scott F, Blanchard J L, Andersen K H. 2014b. Multispecies, trait-based and community size spectrum ecological modelling in R (mizer). Mannual for Mizer, 196

Shannon C E. 1948. A mathematical theory of communication. The Bell System Technical Journal, 27(3): 379–423

Sheldon R W, Prakash A, Sutcliffe Jr W H. 1972. The size distribution of particles in the ocean. Limnology and Oceanography, 17(3): 327–340

Sheldon R W, Sutcliffe Jr W H, Drinkwater K. 1982. Fish production in multispecies fisheries. Canadian Special Publications in Fisheries and Aquatic Sciences, 59: 28–34

Sillero N. 2011. What does ecological modelling model? A proposed classification of ecological niche models based on their underlying methods. Ecological Modelling, 222(8): 1343–1346

Silvert W, Platt T. 1978. Energy flux in the pelagic ecosystem: a time-dependent equation. Limnology and Oceanography, 23(4): 813–816

Sissenwine M P. 1984. Why do fish populations vary?. In: May R M, ed. Exploitation of Marine Communities, Dahlem Workshop Report. Berlin Heidelberg: Springer, 59–94

Smith M D, Fulton E A, Day R W. 2015. Using an Atlantis model of the southern Benguela to explore the response of ecosystem indicators for fisheries management. Environmental Modelling & Software, 69: 23–41

Stockwell D R B, Peterson A T. 2002. Effects of sample size on accuracy of species distribution models. Ecological Modelling, 148(1): 1–13

Svanbäck R, Quevedo M, Olsson J, et al. 2015. Individuals in food webs: the relationships between trophic position, omnivory and among-individual diet variation. Oecologia, 178(1): 103–114

Tang Fenghua, Shen Xinqiang, Wang Yunlong. 2011. Dynamics of fisheries resources near Haizhou Bay waters. Fisheries Science (in Chinese), 30(6): 335–341

Ursin E. 1973. On the prey size preference of cod and dab. Meddelelser fra Danmarks Fiskeri-og Havundersogelser, 7: 85–98

von Foerster H. 1959. Some remarks on changing populations. In: Stohlman F, ed. The Kinetics of Cellular Proliferation. New York: Grune & Stratton, 382–407

Wang Wenhai, Xia Dongxing, Gao Xingchen, et al. 1993. Bays in China: The Bay of Southern Shandong Peninsula and Jiangsu Province (in Chinese). Beijing: China Ocean Press, 448

Werner E E, Gilliam J F. 1984. The ontogenetic niche and species interactions in size-structured populations. Annual Review of Ecology and Systematics, 15(1): 393–425

Whipple S J, Link J S, Garrison L P, et al. 2000. Models of predation and fishing mortality in aquatic ecosystems. Fish and Fisheries, 1(1): 22–40

Ying Yiping, Chen Yong, Lin Longshan, et al. 2011. Risks of ignoring fish population spatial structure in fisheries management. Canadian Journal of Fisheries and Aquatic Sciences, 68(12): 2101–2120

Zhang Chongliang, Chen Yong, Ren Yiping. 2015. Assessing uncertainty of a multispecies size-spectrum model resulting from process and observation errors. ICES Journal of Marine Science, 72(8): 2223–2233